

Ertivo 100mg Film Coated Tablets
Ertivo 150mg Film Coated Tablets

1. NAME OF THE MEDICINAL PRODUCT

Ertivo 100mg Film Coated Tablets
Ertivo 150mg Film Coated Tablets

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Ertivo 100mg Film Coated Tablets: Each tablet contains 109.28mg of Erlotinib Hydrochloride equivalent to 100mg Erlotinib.

Ertivo 150mg Film Coated Tablets: Each tablet contains 163.92mg of Erlotinib Hydrochloride equivalent to 150mg Erlotinib.

For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Film-coated tablet.

Ertivo 100mg Film Coated Tablets: White to yellowish, round, biconvex tablets debossed with 'E' on one side and '100' on other side.

Ertivo 150mg Film Coated Tablets: White to yellowish, round, biconvex tablets debossed with 'E' on one side and '150' on other side.

4. CLINICAL PARTICULARS

4.1 Therapeutic Indications

Non-small cell lung cancer

Ertivo is indicated for the first-line treatment of patients with locally advanced or metastatic non-small cell lung cancer (NSCLC) with EGFR activating mutations.

Ertivo is indicated for maintenance treatment in patients with locally advanced or metastatic NSCLC with EGFR activating mutations who have not progressed after first-line chemotherapy.

Ertivo is also indicated for the treatment of patients with locally advanced or metastatic NSCLC after failure of at least one prior chemotherapy regimen.

Pancreatic Cancer

Ertivo in combination with gemcitabine is indicated for the treatment of patients with locally advanced, unresectable or metastatic pancreatic cancer.

4.2 Posology and Method of Administration

Standard Dosage

Non-small cell lung cancer

EGFR mutation testing should be performed prior to initiation of Ertivo as first-line or maintenance therapy in patients with locally advanced or metastatic NSCLC.

The recommended daily dose of Ertivo is 150 mg taken at least one hour before or two hours after the

ingestion of food.

Pancreatic Cancer

The recommended daily dose of Ertivo is 100 mg taken at least one hour before or two hours after the ingestion of food, in combination with gemcitabine (see the package insert of gemcitabine for the pancreatic cancer indication).

Special Dosage Instructions

Drug Interactions: Concomitant use of CYP 3A4 substrates and modulators may require dose adjustment.

When dose adjustment is necessary, it is recommended to reduce in 50 mg steps.

Hepatic impairment: Erlotinib is eliminated by hepatic metabolism and biliary excretion. Although erlotinib exposure was similar in patients with moderately impaired hepatic function (Child-Pugh score 7-9) compared with patients with adequate hepatic function, caution should be used when administering Ertivo to patients with hepatic impairment. Dose reduction or interruption of Ertivo should be considered if severe adverse reactions occur. Safety and efficacy have not been studied in patients with severe hepatic impairment.

Renal impairment: The safety and efficacy of erlotinib has not been studied in patients with renal impairment.

Pediatric use: The safety and efficacy of erlotinib in the approved indications has not been established in patients under the age of 18 years.

Smokers: Cigarette smoking has been shown to reduce erlotinib exposure by 50-60%. The maximum tolerated dose of erlotinib in NSCLC patients who currently smoke cigarettes was 300 mg. The 300mg dose did not show improved efficacy in second line treatment after failure of chemotherapy compared to the recommended 150mg dose in patients who continue to smoke cigarettes.

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4.3 Contraindications

Ertivo is contraindicated in patients with severe hypersensitivity to erlotinib or to any component of Ertivo.

4.4 Special Warnings and Precautions for Use

Interstitial Lung Disease

Cases of interstitial lung disease (ILD)-like events, including fatalities, have been reported uncommonly in patients receiving erlotinib for treatment of NSCLC (non-small cell lung carcinoma), pancreatic cancer or other advanced solid tumors.

Some examples of reported diagnoses in patients suspected of having ILD-like events include pneumonitis, radiation pneumonitis, hypersensitivity pneumonitis, interstitial pneumonia, interstitial lung disease, obliterative bronchiolitis, pulmonary fibrosis, Acute Respiratory Distress Syndrome, lung infiltration and alveolitis. These ILD-like events started from a few days to several months after initiating erlotinib therapy. Most of the cases were associated with confounding or contributing factors such as concomitant or prior chemotherapy, prior radiotherapy, pre-existing parenchymal lung disease, metastatic lung disease, or pulmonary infections.

In patients who develop acute onset of new and/or progressive unexplained pulmonary symptoms, such as dyspnea, cough and fever, erlotinib therapy should be interrupted pending diagnostic evaluation. If ILD is diagnosed, erlotinib should be discontinued and appropriate treatment initiated as

necessary.

Diarrhea, Dehydration, Electrolyte Imbalance and Renal Failure

Diarrhea has occurred in patients on erlotinib and moderate or severe diarrhea should be treated with loperamide. In some cases, dose reduction may be necessary. In the event of severe or persistent diarrhea, nausea, anorexia, or vomiting associated with dehydration, erlotinib therapy should be interrupted and appropriate measures should be taken to treat the dehydration. There have been rare reports of hypokalemia and renal failure (including fatalities). Some reports of renal failure were secondary to severe dehydration due to diarrhea, vomiting and/or anorexia while others were confounded by concomitant chemotherapy. In more severe or persistent cases of diarrhea, or cases leading to dehydration, particularly in groups of patients with aggravating risk factors (concomitant medications, symptoms or diseases or other predisposing conditions including advanced age), erlotinib therapy should be interrupted and appropriate measures should be taken to intensively rehydrate the patients intravenously. In addition, renal function and serum electrolytes including potassium should be monitored in patients at risk of dehydration.

Hepatitis, Hepatic Failure

Rare cases of hepatic failure (including fatalities) have been reported during use of erlotinib. Confounding factors have included pre-existing liver disease or concomitant hepatotoxic medications. Therefore, in such patients, periodic liver function testing should be considered. Erlotinib dosing should be interrupted if changes in liver function are severe.

Gastrointestinal Perforation

Patients receiving erlotinib are at increased risk of developing gastrointestinal perforation, which was observed uncommonly (including some cases with a fatal outcome). Patients receiving concomitant anti-angiogenic agents, corticosteroids, NSAIDs, and/or taxane based chemotherapy, or who have prior history of peptic ulceration or diverticular disease are at increased risk. Erlotinib should be permanently discontinued in patients who develop gastrointestinal perforation.

Bullous and exfoliative skin disorders

Bullous, blistering and exfoliative skin conditions have been reported, including very rare cases suggestive of Stevens-Johnson syndrome/Toxic epidermal necrolysis, which in some cases were fatal. Erlotinib treatment should be interrupted or discontinued if the patient develops severe bullous, blistering or exfoliating conditions.

Ocular Disorders

Very rare cases of corneal perforation or ulceration have been reported during use of erlotinib. Other ocular disorders including abnormal eyelash growth, keratoconjunctivitis sicca or keratitis have been observed with erlotinib treatment which are also risk factors for corneal perforation/ulceration. Erlotinib therapy should be interrupted or discontinued if patients present with acute/worsening ocular disorders such as eye pain.

Drug Interactions

Erlotinib has a potential for clinically significant drug-drug interactions.

Intolerance to excipients

This medicinal product contains lactose. Patients with rare hereditary problems of galactose intolerance, total lactase deficiency or glucose-galactose malabsorption should not take this medicine. Each Ertivo 100mg and 150mg Film Coated Tablets contain 0.010 mmol (0.24 mg) and 0.016 mmol (0.36 mg) sodium per tablet respectively, equivalent to 0.012% and 0.018% respectively of the WHO recommended maximum daily intake of 2g sodium for adult.

4.5 Interaction with Other Medicinal Products and Other Forms of Interaction

Erlotinib is metabolized in the liver by the hepatic cytochromes in humans, primarily CYP3A4 and to a lesser extent by CYP1A2, and the pulmonary isoform CYP1A1. Potential interactions may occur with drugs which are metabolized by, or are inhibitors or inducers of, these enzymes.

Potent inhibitors of CYP3A4 activity decrease erlotinib metabolism and increase erlotinib plasma concentrations. Inhibition of CYP3A4 metabolism by ketoconazole (200 mg po BID for 5 days) resulted in increased exposure to erlotinib.

When erlotinib was co-administered with ciprofloxacin, an inhibitor of both CYP3A4 and CYP1A2, the erlotinib exposure [AUC] and maximum concentration ($C_{\max}$) increased by 39% and 17%, respectively. Therefore caution should be used when administering erlotinib with potent CYP3A4 or combined CYP3A4/CYP1A2 inhibitors. In these situations, the dose of erlotinib should be reduced if toxicity is observed.

Potent inducers of CYP3A4 activity increase erlotinib metabolism and significantly decrease erlotinib plasma concentrations.

For patients who require concomitant treatment with erlotinib and a potent CYP3A4 inducer such as rifampicin an increase in dose to 300 mg should be considered while their safety is closely monitored, and if well tolerated for more than 2 weeks, further increase to 450 mg could be considered with close safety monitoring. Higher doses have not been studied in this setting.

Pre-treatment or co-administration of erlotinib did not alter the clearance of the prototypical CYP3A4 substrates midazolam and erythromycin. Significant interactions with the clearance of other CYP3A4 substrates are therefore unlikely. Oral availability of midazolam did appear to decrease by up to 24%, which was however not attributed to effects on CYP3A4 activity.

The solubility of erlotinib is pH dependent. Erlotinib solubility decreases as pH increases. Drugs that alter the pH of the upper GI tract may alter the solubility of erlotinib and hence its bioavailability. Co-administration of erlotinib with omeprazole, a proton pump inhibitor, decreased the erlotinib exposure [AUC] and $C_{\max}$ by 46% and 61%, respectively. There was no change to $T_{\max}$ or half-life. Concomitant administration of erlotinib with 300 mg ranitidine, an H₂-receptor antagonist, decreased erlotinib exposure [AUC] and $C_{\max}$ by 33% and 54%, respectively. Therefore, co-administration of drugs reducing gastric acid production with erlotinib should be avoided where possible. Increasing the dose of erlotinib when co-administered with such agents is not likely to compensate for this loss of exposure. If patients need to be treated with such drugs, then an H₂-receptor antagonist such as ranitidine should be considered and used in a staggered manner. Erlotinib must be taken at least 2 hours before or 10 hours after the H₂-receptor antagonist dosing.

Interaction with coumarin-derived anticoagulants, including warfarin, leading to increased International Normalized Ratio (INR) and bleeding events, which in some cases were fatal have been reported in patients receiving erlotinib. Patients taking coumarin-derived anticoagulants should be monitored regularly for any changes in prothrombin time or INR.

The combination of erlotinib and a statin may increase the potential for statin-induced myopathy, including rhabdomyolysis, which was observed rarely.

Smokers should be advised to stop smoking as cigarettes smoking, which is known to induce CYP1A1 and CYP1A2, has been shown to reduce erlotinib exposure.

In a study, there were no significant effects of gemcitabine on the pharmacokinetics of erlotinib nor were there significant effects of erlotinib on the pharmacokinetics of gemcitabine.

4.6 Fertility, pregnancy and lactation

Potential Females and Males of Reproductive Potential

Contraception: Females:

Adequate contraceptive methods should be used during therapy, and for at least 2 weeks after completing therapy.

Pregnancy

There are no adequate or well controlled studies in pregnant women using erlotinib. Studies in animals have shown some reproductive toxicity. The potential risk for humans is unknown. Women of childbearing potential must be advised to avoid pregnancy while on erlotinib. Treatment should only be continued in pregnant women if the potential benefit to the mother outweighs the risk to the fetus.

Lactation

Nursing mothers: It is not known whether erlotinib is excreted in human milk. No studies have been conducted to assess the impact of erlotinib on milk production or its presence in breast milk. As the potential for harm to the nursing infant is unknown, mothers should be advised against breastfeeding while receiving erlotinib and for at least 2 weeks after the final dose.

4.7 Effects on Ability to Drive and Use Machines

Erlotinib has no or negligible influence on the ability to drive and use machines.

4.8 Undesirable Effects

The incidence of ADRs from clinical trials and the post marketing setting reported with erlotinib alone or in combination with chemotherapy are summarised in the table below. Adverse drug reactions are listed by MedDRA system organ class. The corresponding frequency category for each adverse drug reaction is based on the following convention: very common ($\geq 1/10$), common ($\geq 1/100$ to $< 1/10$), uncommon ($\geq 1/1,000$ to $< 1/100$), rare ($\geq 1/10,000$ to $< 1/1,000$), very rare ($< 1/10,000$), not known (cannot be estimated from the available data).

Within each frequency grouping, adverse reactions are presented in order of decreasing seriousness.

Table 1: Adverse Drug Reactions

Infections and infestations	
Very Common	Infection*
Metabolism and nutrition disorders	
Very Common	Anorexia, Weight decreased
Psychiatric disorders	
Very Common	Depression
Nervous system disorders	
Very Common	Headache, Neuropathy
Eye disorders	
Very Common	Conjunctivitis, Keratoconjunctivitis sicca
Common	keratitis, conjunctivitis
Uncommon	eyelash changes*
Very rare	corneal perforations, corneal ulcerations, uveitis
Respiratory, thoracic and mediastinal disorders	
Very Common	Dyspnea, Cough
Common	Epistaxis
Uncommon	interstitial lung disease*
Gastrointestinal disorders	
Very Common	Diarrhea*, Nausea, Vomiting, Stomatitis, Abdominal pain, Dyspepsia,

	Flatulence
Common	gastro-intestinal bleeding*
Uncommon	gastro-intestinal perforations*
Rare	pneumatosis intestinalis
Hepatobiliary disorders	
Very common	liver function test abnormalities*
Rare	hepatic failure*, hepatitis
Not known (Cannot be estimated from the available data)	acute hepatitis
Skin and subcutaneous tissue disorders	
Very Common	Rash*, Pruritus
Common	alopecia, dry skin, paronychia, folliculitis, acne/dermatitis acneiform, skin fissures
Uncommon	hirsutism, eyebrow changes, brittle and loose nails, mild skin reactions such as hyperpigmentation
Rare	palmar plantar erythrodysaesthesia syndrome
Very rare	Stevens-Johnson syndrome/Toxic epidermal necrolysis*
Renal and urinary disorders	
Common	renal insufficiency
Uncommon	nephritis, proteinuria
General disorders & administration site conditions	
Very Common	Fatigue, Pyrexia, Rigors

* For additional details, see section "Description of selected adverse reactions" below

Description of selected adverse reactions

Rash

Rash includes dermatitis acneiform. In general, rash manifests as a mild or moderate erythematous and papulopustular rash, which may occur or worsen in sun exposed areas. For patients who are exposed to sun, protective clothing, and/or use of sunscreen (e.g. mineral-containing) may be advisable.

Diarrhoea

Diarrhoea can lead to dehydration, hypokalemia and renal failure. It includes fatalities.

Infection

This can be severe infections with or without neutropenia, including pneumonia, sepsis, and cellulitis.

Eyelash changes

Changes include in-growing eyelashes, excessive growth and thickening of the eyelashes.

Interstitial lung disease (ILD)

ILD includes fatalities in patients receiving erlotinib for treatment of NSCLC or other advanced solid tumours. A higher incidence has been observed in patients in Japan.

Gastro-intestinal (GI) bleeding

GI bleeding includes fatalities. In clinical studies, some cases have been associated with concomitant warfarin administration and some with concomitant NSAID administration. Gastro-intestinal perforations also include fatalities.

Liver function test abnormalities

Abnormalities include increased alanine aminotransferase [ALT], aspartate aminotransferase [AST] and bilirubin. Cases were mainly mild to moderate in severity, transient in nature or associated with

liver metastases.

Hepatic failure

This includes fatalities. Risk factors may include pre-existing liver disease or concomitant hepatotoxic medications.

Stevens-Johnson syndrome/Toxic epidermal necrolysis

This includes fatalities.

4.9 Overdose

Severe adverse events such as diarrhea, rash, and possibly liver transaminase elevation may occur above the recommended dose of 150 mg. In case of suspected overdose erlotinib should be withheld and symptomatic treatment administered.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic Properties

Mechanism of Action

Erlotinib potently inhibits the intracellular phosphorylation of HER1/EGFR receptor. HER1/EGFR receptor is expressed on the cell surface of normal cells and cancer cells. In non-clinical models, inhibition of EGFR phosphotyrosine results in cell stasis and/or death.

5.2 Pharmacokinetic Properties

Exposure

Following a 150 mg oral dose of erlotinib, at steady state, the median time to reach maximum plasma concentrations is approximately 4.0 hours with median maximum plasma concentrations achieved of 1,995 ng/ml. Prior to the next dose at 24 hours, the median minimum plasma concentrations are 1,238 ng/ml. Median AUC achieved during the dosing interval at steady state are 41.300 mcg*hr/ml.

Absorption

Oral erlotinib is well absorbed and has an extended absorption phase, with mean peak plasma levels occurring at 4 hours after oral dosing. A study in normal healthy volunteers provided an estimate of bioavailability of 59%. The exposure after an oral dose may be increased by food.

Following absorption, erlotinib is highly bound in blood, with approximately 95% bound to blood components, primarily to plasma proteins (i.e. albumin and alpha-1 acid glycoprotein [AAG]), with a free fraction of approximately 5%.

Distribution

Erlotinib has a mean apparent volume of distribution of 232 L and distributes into tumor tissue of humans.

Metabolism

Erlotinib is metabolised in humans by hepatic cytochrome P450 enzymes, primarily by CYP3A4 and to a lesser extent by CYP1A2. Extrahepatic metabolism by CYP3A4 in intestine, CYP1A1 in lung, and CYP1B1 in tumour tissue potentially contribute to the metabolic clearance of erlotinib. There are three main metabolic pathways identified: 1) O-demethylation of either side chain or both, followed by oxidation to the carboxylic acids; 2) oxidation of the acetylene moiety followed by hydrolysis to the aryl carboxylic acid; and 3) aromatic hydroxylation of the phenyl-acetylene moiety. The primary metabolites of erlotinib produced by O-demethylation of either side chain have comparable potency to erlotinib in nonclinical *in vitro* assays and *in vivo* tumor models. They are present in plasma at levels that are <10%

of erlotinib and display similar pharmacokinetics as erlotinib.

Elimination

The metabolites and trace amounts of erlotinib are excreted predominantly via the feces (>90%), with renal elimination accounting for only a small amount of an oral dose.

Clearance

A study of patients receiving single agent erlotinib shows a mean apparent clearance of 4.47 L/hour with a median half-life of 36.2 hours. Therefore, the time to reach steady state plasma concentration would be expected to occur in approximately 7-8 days. No significant relationships between predicted apparent clearance and patient age, body weight, gender, and ethnicity were observed.

Patient factors, which correlate with erlotinib pharmacokinetics, are serum total bilirubin, AAG concentrations and current smoking. Increased serum concentrations of total bilirubin and AAG concentrations were associated with a slower rate of erlotinib clearance. Smokers had a higher rate of erlotinib clearance.

Co-administration of gemcitabine had no effect on erlotinib plasma clearance.

Pharmacokinetics in Special Populations

There have been no specific studies in pediatric or elderly patients.

Hepatic impairment

Erlotinib is mainly cleared by the liver. Erlotinib exposure was similar in patients with moderately impaired hepatic function (Child-Pugh score 7-9) compared with patients with adequate hepatic function including patients with primary liver cancer or hepatic metastases.

Renal impairment

Erlotinib and its metabolites are not significantly excreted by the kidneys, as less than 9% of a single dose is excreted in the urine. No clinical studies have been conducted in patients with compromised renal function.

Smokers

A study in nonsmoking and currently cigarette smoking healthy subjects has shown that cigarette smoking leads to increased clearance of, and decreased exposure to, erlotinib. This reduced exposure in current smokers is presumably due to induction of CYP1A1 in lung and CYP1A2 in the liver.

6 PHARMACEUTICAL PARTICULARS

6.1 List of Excipients

Tablet core:

Microcrystalline Cellulose, Lactose Monohydrate, Hydroxypropyl Cellulose, Sodium Lauryl Sulfate, Crospovidone, Colloidal Silicon Dioxide, Magnesium Stearate

Film coat:

Opadry Complete Film Coating System Y-5-7068 White (100mg and 150mg)

6.2 Incompatibilities

Not applicable

6.3 Shelf Life

Please refer to the expiry date on the product labels.

6.4 Special Precautions for Storage

Store below 30°C.

6.5 Nature and Content of Container

Ertivo 100mg Film Coated Tablets: Alu/Alu blister of 1x10's, 3x10's, 6x10's, 10x10's, 50x10's and 100x10's tablets packed in the outer carton. Not all pack size will be marketed.

Ertivo 150mg Film Coated Tablets: Alu/Alu blister of 1x10's, 3x10's, 6x10's, 10x10's, 50x10's and 100x10's tablets packed in the outer carton. Not all pack size will be marketed.

6.6 Special precautions for disposal and other handling

No special requirements.

7 MANUFACTURER

Manufactured by:

Novugen Oncology Sdn. Bhd.
No. 47, Jalan Lengkok Teknologi 2,
Taman Teknologi Enstek Fasa 1,
71760 Bandar Baru Enstek,
Negeri Sembilan, Malaysia.

Product Registration Holder:

Novugen Oncology Sdn. Bhd.
No. 3, Jalan Jururancang U1/21,
Hicom Glenmarie Industrial Park,
40150 Shah Alam, Selangor, Malaysia

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8 DATE OF REVISION

19/02/2024

Package insert is drafted in Times New Roman with font size of 11, subject to change based on any other legible font and font size.
